

DCR3: The Nominator Correctly Identifies the Load-Bearing Commitment. It Isn't the One Einstein Deleted.

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Date-Cut Retrodiction – DCR3 (nominator on real material) **Status:** Overall **NO_GO**. M1 NO_GO (T1 ranks 3rd at the 1904 target cut, behind unclassified and T2_privileged_frame). M3 auto-NO_GO because M1 failed. M2 GO. M4 GO. **The DR-arc's execution-free nominator does not solve its own load-bearing question on real material – and the specific way it fails is illuminating.** **Date:** 2026-07-27

Abstract

DCR1–DCR2b built the extraction, matching, and class-aggregation pipeline. DR5/DR7 proved a structural wall for proposition-ranking nominators on multi-realisation commitments. DCR3 is the DR-arc's original load-bearing question: given only propositions extracted from pre-1905 material, does the DR-arc's execution-free scoring function rank the T1 class (Einstein's actual deletion) *first* at the 1904 target cut, at a level that beats random?

Preregistered scoring function: $\text{score}(p) = \text{kind_weight}(p) * \text{degree}(p)$, where kind_weight is 3 / 2 / 1 for required_by_argument / presupposed / asserted (DR-arc precedent), and degree = number of documents at the cut that share content-stem-equivalent propositions (identical Jaccard threshold to consensus.py). $\text{Class-level} = \text{sum of member scores with multidoc}(\text{min_docs}=2) \text{ gating from DCR2b.}$

Result at 1904:

rank	class	score	n members	n documents
1	unclassified	302	235	15
2	T2_privileged_frame	15	11	8
3	T1_absolute_simultaneity	9	7	5
4	T3_local_time_artifice	2	2	2

M1 NO_GO. T1 ranks third. Random-null probability of T1 first = 0.25 (uniform over 4 classes); the scored ranking did not clear it either. M3 NO_GO by construction (M1 failed). M2 GO (T1 also not first at 1880). M4 GO (scoring function committed, digest recorded).

Overall NO_GO. The DR-arc program's execution-free scoring cannot identify Einstein's deletion on real material.

But NOT because the scoring function is broken. Because "*load-bearing in the corpus*" and "*deletable in a scientific revolution*" are different concepts, and Einstein's move required exactly the second. The nominator's high scores go to the aether frame (T2, 15) – a commitment stated widely across the electrodynamics literature – and to the huge unclassified bucket. The commitment Einstein actually deleted (T1) scored low precisely because it was *rarely stated*. Rarely stated meant nobody was defending it, which is what made it available to be given up.

The load-bearing-in-text ↔ deletable-in-revolution asymmetry is not a bug in the algorithm. It's a substantive finding about how conceptual revolutions actually work, and it converts DR5/DR7's structural wall from an abstract theorem into a specific empirical observation on a case with ground truth attached.

1. What was preregistered

DCR3_PREREGISTRATION.md (2026-07-27, before nominate_dcr3.py was drafted or scored) fixed:

kind_weight = 3 for required_by_argument, 2 for presupposed, 1 for asserted (an ordinal encoding of the DR-arc's stated preference for presuppositional commitments). degree = number of distinct documents at the cut with content-stem-equivalent propositions (identical Jaccard equivalence rule to consensus.py, threshold 0.5).

(M1 result beats random null at $p < 0.01$), M4 (scoring function committed by SHA-256 digest).

- Corpus: DCR1e consensus at each cut (byte-identical to published).
- Class assignment: target_v4 from DCR1f (byte-identical).
- Aggregation rule: multidoc(min_docs=2) from DCR2b.
- Scoring function: $\text{score}(p) = \text{kind_weight}(p) * \text{degree}(p)$.
- Ground truth: T1 class (per DCR1c/d/e/f/2a/2b).
- Baseline: 10,000 random permutations of class keys (uniform null).
- Four gates: M1 (T1 first at 1904), M2 (T1 not first at 1880), M3

Nothing tuned after results. nominate_dcr3.py SHA-256:
bef69991ee199e32746acd994fbab7dab61ee6d238939d0fc4438d967664209b.

2. Results at each cut

1904 (target cut, 15 documents, 254 consensus propositions):

rank	class	score	n members	n documents	mean per-member
1	unclassified	302	235	15	1.28
2	T2_privileged_frame	15	11	8	1.36
3	T1_absolute_simultaneity	9	7	5	1.29
4	T3_local_time_artifice	2	2	2	1.00

1880 (deep placebo cut, 3 documents, 51 consensus propositions):

rank	class	score	n members
1	unclassified	42	50
2 (0-score tie)	T1_absolute_simultaneity	0	1 (multidoc-demoted, single doc)
3 (0-score tie)	T2_privileged_frame	0	0
4 (0-score tie)	T3_local_time_artifice	0	0

1897 (near placebo, 8 documents, 135 consensus propositions):

Same shape as 1904 – unclassified dominates, T2 wins among semantic classes (score 8), T1 third (score 2), T3 zero.

3. Gate decisions

gate		
M1 T1 first at 1904	NO_GO	T1 rank 3, behind unclassified and T2
M2 T1 not first at 1880	GO	multidoc de-ranks T1 to score 0
M3 beats random null	NO_GO	M1 failed; null $p = 0.25$
M4 scoring function committed	GO	SHA-256 recorded and pinned

Overall NO_GO on the preregistered DR-arc question.

Licensed reading (from the runner):

DR_arc_does_not_solve_its_load_bearing_question: T1 does not rank first at 1904 under the preregistered execution-free scoring. Combined with DR5/DR7 predictions, this is a clean structural falsification: the DR-arc as designed cannot identify multi-realisation commitments even with correct class grouping and placebo-clean aggregation.

4. The substantive finding under the null

The nominator did not fail in the "buggy" sense. It correctly identified the commitments that are load-bearing across the pre-1905 electrodynamics corpus:

documents. It is dominated by ordinary physical assertions – Maxwell's field equations, energy propagation, ether elasticity, refraction data. These are load-bearing in the specific sense that every derivation and every experiment in the corpus references them. They are *not* candidates for deletion; they are the shared background against which any specific commitment gets stated.

Wide coverage: Michelson, Michelson-Morley, FitzGerald, Larmor, Lodge, Rayleigh, Brace, Lorentz all invoke it. It was the *actively disputed* commitment of the period – the null-result experiments (MM 1887) were arguing about *whether the aether is at rest with respect to the earth or moves through it*. Lorentz's 1904 paper provided a fully-worked accommodation. T2 was under maximum argumentative pressure and everyone knew about it.

because absolute simultaneity was less important – because it was less *discussed*. Newton stated it in 1687; by 1900 it had gone silent, exactly the pattern DCR1d identified. Only Poincaré 1898 named it, and only as a philosophical curiosity ("the measure of time"), not as a physical commitment under revision.

- **The unclassified bucket** contains 235 propositions across 15
- **T2 (privileged aether frame)** scored 15 across 8 documents.
- **T1 (absolute simultaneity)** scored 9 across 5 documents. Not

The DR-arc's scoring rule assumed: **most load-bearing → most deletable**. That assumption is false on this case. Einstein's move required identifying a commitment that was **silent in the literature because it was universally presupposed** – the opposite of the "widely-defended" pattern that scores high. The nominator correctly identified the commitments that were under revision (T2, the aether frame). But Einstein did not delete T2 – Lorentz and others were already producing theories that preserved much of it. Einstein deleted T1, the commitment nobody had thought to argue about.

This asymmetry is the finding. It converts DR5/DR7's structural wall from an abstract theorem into an empirical observation with teeth: on a real conceptual-change case with ground truth, an execution-free scoring function based on textual load-bearing-ness correctly identifies the wrong commitment. The load-bearing commitment (T2) is not the deletable one. The deletable one (T1) is by construction the one that scores low.

5. What DCR3 licenses

NO_GO on the DR-arc program as designed for this class of commitments. Any future DR-arc paper attempting to rank "deletable commitments" from textual load-bearing-ness has to answer DCR3's specific finding first: revolutions delete the silent commitments, not the loud ones.

A specific consequence: **class-based nomination that scores by "widely stated and defended" will systematically miss the revolutionary deletions.** For scoring to reach silent-but-presupposed commitments, the scorer needs access to something other than corpus frequency – either semantic reasoning about what each argument *requires* (DR5's condition (b) semantic access, which requires an LLM), or an oracle for "which commitments were presupposed but not

stated" (DR7's grouping function correctness, open problem).

Two research paths remain open:

score for "how presuppositional is this commitment given the corpus's arguments." DR6/DR6e showed LLM semantic access escapes DR5's wall for accessible D; this would test it for the T1 target. Not a proposition-ranking nominator any more; the LLM does the ranking.

proposition-ranking scorer based on corpus statistics cannot identify silent-but-presupposed commitments. Would generalise the DR3 empirical finding into a specific corollary of DR5/DR7.

- **Semantic-scoring path** – replace degree with a Claude-based
- **Theorem-shape path** – prove formally that any

6. What DCR3 does not license

proposition-ranking scorer can distinguish D from its realisations, and the wall's severity depends on whether the scorer has access to something beyond surface statistics. The nominator scored surface load-bearing-ness. The wall bit.

question – the pipeline is honest, the gates are honored, and the program can now report a specific empirical result about how the wall shows up on a real case with ground truth. Failing the load-bearing gate honestly is exactly what a working program looks like when it hits the limit it was structured to detect.

that a *specific class* of scoring functions cannot reach T1. A semantic-scoring approach (DR-6-style LLM verifier) may or may not reach it; DCR3 does not test that.

- **DR5 is refuted.** It isn't. DR5 predicted exactly this: no
- **The DR-arc has failed.** It has succeeded at its structural
- **Einstein's move is inaccessible in principle.** DCR3 shows only

7. Historical corollary

There is an implicit claim about scientific revolutions in DCR3's finding that is worth surfacing. Revolutions delete the commitments that had gone silent – not the ones under argument. This has echoes in the philosophy-of-science literature (Kuhn's notion of tacit knowledge; Lakatos's protective belt) but the specific mechanism here is sharper: the deletion becomes *available* only when the commitment stops being defended, because defended commitments have a community actively arguing for them and unrelated ones don't attract the deletion move. This mechanism is quantifiable – the DR-arc's scoring rule is essentially a "how defended is this?" metric, and it correctly identifies T2 as more defended than T1. The revolution picked the *undefended* one.

If replicable across other conceptual-change cases (Copernicus, Darwin, Lavoisier – each with a known deleted commitment), this would be a substantive philosophy-of-science claim testable with DCR-arc methodology.

Appendix: reproduction

```
uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr3
```

Deterministic (fixed seed 20260727 for null permutations). Writes results/dcr3_verdict.json. Local CPU, seconds.

Preregistration **digest** **(SHA-256** **of** **DCR3_PREREGISTRATION.md):**
263af1981bcf298fa28d60e2aa599d8d96decdd9e4f77d3a5b7363e27861d4c2.

Scoring **module** **digest** **(SHA-256** **of** **nominate_dcr3.py):**
bef69991ee199e32746acd994fbab7dab61ee6d238939d0fc4438d967664209b.